Lab 3: Explaining Your Detectors and Building a Belief Rule-Based Expert System

1. What is this lab about?

A good detector is not enough if nobody can see why it made a decision. In this lab you open up the detectors you already trained, explain them with **SHAP** and **LIME**, and then turn the most important features into a **Belief Rule-Based Expert System (BRBES)**: a small set of readable rules where each rule says how sure it is, and can even admit when it does not know.

You will run the same pipeline for more than one **learning paradigm** and compare them:

- **Supervised** — your Lab 1 model (required).
- **Semi-supervised** — your Lab 2 model (required).
- **Self-supervised** — a simple self-trained model, if you have time (optional bonus).

For each model you will explain it, pick its two most important features, and build a BRBES from them.

Please note that this Lab to cover (Tuesday lecture : Knowledge based system for cyber defense and Thursday lecture: Explainable AI: Principles and methods)

2. What you will be able to do afterwards

- Explain a trained model's decisions with SHAP and LIME.
- Judge explanations on faithfulness, stability, and readability.
- Turn feature importance into a small, transparent belief rule base.
- Run the belief-rule inference (input transformation, activation weights, evidential reasoning) and read the belief output.
- Compare explainability and accuracy across supervised, semi-supervised, and self-supervised models.

3. Tools and setup

- **Language:** Python 3. Work in a Jupyter notebook or google colab or plain scripts , you can use what you prefer.
- **Core libraries:** pandas, scikit-learn, matplotlib, plus shap and lime for the explanations.
- **Reuse Labs 1 and 2:** load the models you already trained and the same dataset and test set.
- **Install once:** pip install shap lime (add your Lab 1 and Lab 2 libraries if a fresh setup).

4. The models and data

Use the **same dataset and test set as Labs 1 and 2** (CICIDS2017 or CICIDS2018), so every result is comparable. Each model below is treated as a “black box” that you will explain.

Paradigm	Model to use	Required?
Supervised	Your trained Lab 1 detector.	Yes
Semi-supervised	Your trained Lab 2 detector (pseudo-labelling or the method you chose).	Yes
Self-supervised	A simple model you train without labels, for example an autoencoder on the unlabelled data, then a small classifier on top.	Optional

Make life easier
SHAP and LIME can be slow on big data. Use a smaller, balanced sample for the explanations. Smaller data also makes your rules easier to read.

What a "black box" is (and why we explain it)
Your models give good predictions but not reasons. SHAP and LIME add the reasons, so an analyst can see why an alert was raised and decide whether to trust it. A BRBES goes further: it is not a black box at all, every step can be shown to the analyst.

5. Methods you will use

Use **SHAP** and **LIME** to explain each model, then build a **Belief Rule-Based Expert System** on its two most important features.

Method	What it is / beginner tip
SHAP	Gives each feature a fair share of a prediction, based on game theory. Shows both the whole-model view (which features matter most) and single-sample views. Good for choosing the top features.
LIME	Fits a simple, readable model around one prediction to show which features pushed it toward normal or attack. Works one sample at a time. Good as a check on SHAP.
Belief Rule-Based Expert System (BRBES)	A small set of IF-THEN rules. Each rule's THEN part is a belief distribution over risk levels (Low, Medium, High), not a single label. Fully readable, and it can say how sure it is, or that it does not know.

Comparing explanations (what to look for)
Faithfulness: does the explanation match how the model really behaves?
Stability: run it twice on the same sample, do you get the same explanation?
Readability: how easy is it for a person to understand?

The belief-rule framework (the five steps you will implement)
1. Knowledge representation: write the rules, each with belief degrees (β), a rule weight (θ), and attribute weights (δ).
2. Input transformation: turn each crisp feature value into matching degrees over its levels.
3. Rule activation weight: combine the matching degrees with θ and δ, then normalise, to see which rules matter.

4. Belief update: if a feature is missing, shrink that rule's belief; the shortfall is kept as ignorance, not hidden.

5. Rule aggregation: combine the active rules with the Evidential Reasoning (ER) approach into one belief distribution.

6. Step-by-step tasks

Step 1: Gather your models and data

- Load your Lab 1 (supervised) and Lab 2 (semi-supervised) models, and the same dataset and test set. Optionally train a small self-supervised model (for example an autoencoder on the unlabelled data, then a small classifier).

Step 2: Explain each model with SHAP

- Run SHAP on each model. Make a summary plot for the whole model (which features matter most) and a waterfall or force plot for two or three single predictions.

Step 3: Explain each model with LIME

- Run LIME on the same predictions for each model. Note which features it says pushed the decision toward attack or normal.

Step 4: Compare the explanations

- For each model, judge SHAP and LIME on faithfulness, stability, and readability (see the note above). Say where they agree and where they do not.

Step 5: Choose the top 2 features for each model

- From the SHAP whole-model importance, checked against LIME, pick the two most important features for each model. The paradigms may pick different features, that itself is a finding, so write it down.

Step 6: Build a belief rule base for each model

- For each of the two features, define levels such as Low, Medium, High (for example, from data quantiles). Choose levels for the output too: attack risk Low, Medium, High.
- Make one rule for each combination of levels (up to 2 x 2x 2 = 8 rules). Give each rule a belief distribution over {Low, Medium, High} risk, a rule weight θ, and attribute weights δ. Set the beliefs from the data (the share of attacks in that group) or from expert judgment.

What a belief rule looks like

IF Feature A is HIGH AND Feature B is HIGH THEN attack risk { High 0.85, Medium 0.15, Low 0.00 }, with rule weight θ = 1 and attribute weights δ = (1, 1, 1).

The three numbers are belief degrees. They show how sure the rule is. If they add up to less than 1, the remainder is honest ignorance, the rule's way of saying it does not fully know.

Step 7: Run the expert system

- Input transformation: turn each input value into degrees over its levels (a value between Medium and High counts partly as each).
- Rule activation weight: work out how strongly each rule matches, using θ and δ, then normalise so the weights sum to 1.

- Belief update: if a feature is missing for a sample, shrink that rule's belief and keep the shortfall as ignorance.
- Rule aggregation: combine the active rules with the Evidential Reasoning (ER) approach into one belief over {Low, Medium, High}.
- Decision: turn the belief into a utility score (for example Low 0, Medium 50, High 100) and label the sample as Attack if the score passes your threshold.

Building the BRBES

You may implement the belief-rule inference yourself in Python (it is not long, just follow the five steps) or use an existing RIMER / ER implementation. Either way, cite what you used. If you have time, you can tune θ, δ and β with a genetic algorithm, the rules stay readable afterwards.

Step 8: Test and compare

- Score each BRBES on the same test set (accuracy, macro-F1, recall, ROC-AUC, FAR).
- Compare in two directions: each BRBES against its own black-box model, and the BRBES across paradigms (supervised vs semi-supervised vs self-supervised).
- Discuss: do the paradigms rely on the same top features? Which BRBES is most accurate, and is the drop from the black box worth the extra transparency?
- **Right metric:** because the data is imbalanced, judge with macro-F1, recall and FAR, not accuracy alone.

7. What to hand in

Your code. A notebook or scripts plus a short README or PDF that lists the libraries and how to run it. Include the random seed and your Lab 1 and Lab 2 models.

A short report (2–3 pages). Suggested sections:

- What the problem is (explaining a model, and rule-based reasoning across paradigms) — a short paragraph.
- What you did (SHAP and LIME, the top 2 features per model, the BRBES you built).
- Results (your explanation comparison, a few example rules with belief degrees, and a metrics table for each BRBES vs its black box).
- Comparison across paradigms (which features each one used, and which BRBES did best).
- Discussion (which explanations you trust, and the accuracy vs explainability trade-off). Who did what (one or two lines).

How your work is graded

Every lab is graded 3 / 4 / 5 / F on the same four things. You do not need to be perfect to pass , you need the work to run, to make sense, and to be clearly reported. The table shows what “good” looks like for this lab.

Criterion	Weight	What good work looks like here
Implementation and code quality	30%	SHAP and LIME run and reproduce; the BRBES is implemented and runs the five belief-rule steps; reuses your Lab 1 and Lab 2 models; README; seed is fixed.

Criterion	Weight	What good work looks like here
Experimental design and methodology	25%	Explanations compared on faithfulness, stability and readability; top-3 features chosen per model with a reason; belief rules, weights and ER used correctly; supervised and semi-supervised done (self-supervised optional); same test set.
Results and analysis	20%	Explanation comparison reported; example rules with belief degrees shown; BRBES metrics vs black box for each paradigm; the paradigms compared; the trade-off discussed.
Report quality	25%	Clear 2–3 pages; captioned figures/tables used in the text; who-did-what line included+Appendix: screenshot of your code with comment on each part for what you used.

Submitting your work

- **Where.** Upload to CANVAS by the deadline: your code (a zip or a repository link) and the report as a PDF. Late submissions are not accepted in general.
- **Group work.** Work together for real. Add a short line at the end of the report saying who did what. If one person does everything, that shows up in individual grades.
- **Second chance.** If a lab does not reach grade 3 you may resubmit it once, before the final reporting date in the schedule.
- **Honesty.** You may use libraries, tutorials, and public code , just cite them. If you use an AI assistant, say so in the report. Your results must come from your own submitted code.

Common mistakes to avoid

- Building only one BRBES, you need at least supervised and semi-supervised to compare.
- Trusting one tool blindly, SHAP and LIME can disagree, so look at both.
- Picking the tt features by guess instead of from the SHAP and LIME results.
- Setting every belief degree to 0 or 1, which throws away the belief and the ignorance that make a BRBES useful.
- Forgetting the utility score and threshold when turning a belief distribution into a Normal / Attack decision.

Where to read more

- Belief rules and the five-step inference: your course lecture “Expert Systems and Rule-Based Reasoning for Cyber Defense” (Prof. M. Shahadat Hossain).
- BRBES / RIMER: Yang, J.-B., Liu, J., Wang, J., Sii, H.-S., & Wang, H.-W. (2006). Belief rule-base inference methodology using the evidential reasoning approach - RIMER. IEEE Transactions on Systems, Man, and Cybernetics - Part A, 36(2), 266-285.
- SHAP: Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 30.
- LIME: Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why Should I Trust You?”: Explaining the Predictions of Any Classifier. KDD ’16, 1135-1144.
- Dataset (same as Labs 1 and 2)